

StreamWorks Media – Customer Churn Analysis

Introduction

As part of the Data Strategy Team at StreamWorks Media, this project focused on understanding customer churn within an increasingly competitive streaming industry dominated by platforms such as Netflix and Amazon Prime Video.

With rising customer acquisition costs, retaining existing subscribers has become essential for sustainable growth and profitability. This analysis aimed to identify the behavioural patterns and business factors contributing to customer churn while exploring opportunities to improve subscriber retention.

Using data cleaning, feature engineering, statistical testing, and predictive modelling, the project evaluated how customer engagement, loyalty, promotions, and platform usage influence churn risk. The insights generated help support targeted retention strategies and improve long-term customer value.

Preparing the Data for Analysis

Before analysing customer behaviour, the dataset required extensive cleaning and preparation to ensure consistency and reliability.

Date-related columns such as signup date and last active date were converted into datetime format using Pandas to support time-based analysis. Rows containing critical missing values including customer IDs, churn status, and activity dates were removed to maintain data integrity.

	user_id	age	gender	signup_date	last_active_date	country	\
0	1001.0	56.0	2	2025-02-04	2025-07-13	1	
1	1002.0	69.0	1	2023-02-01	2025-07-13	3	
2	1003.0	46.0	1	2022-08-21	2025-07-13	4	
3	1004.0	32.0	2	2023-09-14	2025-07-13	2	
4	1005.0	60.0	0	2023-07-29	2025-07-13	3	
	subscription_type	average_watch_hours	mobile_app_usage_pct	\			
0	2	42.6	77.4				
1	0	65.3	98.0				
2	1	40.1	47.8				
3	1	5.8	53.2				
4	2	32.7	16.8				
	complaints_raised	received_promotions	referred_by_friend	is_churned	\		
0	1.0	0	0	1.0			
1	4.0	0	1	1.0			
2	0.0	0	1	1.0			
3	1.0	1	1	1.0			
4	5.0	0	1	0.0			
	monthly_fee	tenure_days	is_loyal				
0	10.99	159.0	False				
1	5.99	893.0	True				
2	13.99	1057.0	True				
3	13.99	668.0	True				
4	9.99	715.0	True				

Missing numerical values were handled using median imputation through the `.fillna()` method, reducing the impact of outliers while preserving the dataset structure.

To improve model usability, categorical variables were encoded using Label Encoding across object-type columns. This enabled machine learning models to process behavioural and demographic features efficiently.

Building Features to Understand Customer Behaviour

To better explain churn behaviour, several new analytical features were engineered from the raw dataset.

Tenure Days

Measured the total duration of a customer's subscription, helping identify whether newer subscribers were more likely to churn.

Loyalty Indicator

Flagged customers who remained subscribed beyond a defined threshold, allowing analysis of long-term retention behaviour.

Watch-to-Fee Ratio

Compared customer viewing hours against subscription cost to estimate perceived customer value.

Heavy Mobile User

Identified customers who primarily consumed content on mobile devices.

Promotional & Engagement Features

	user_id	age	gender	signup_date	last_active_date	country	\
0	1001.0	56.0	2	2025-02-04	2025-07-13	1	
1	1002.0	69.0	1	2023-02-01	2025-07-13	3	
2	1003.0	46.0	1	2022-08-21	2025-07-13	4	
3	1004.0	32.0	2	2023-09-14	2025-07-13	2	
4	1005.0	60.0	0	2023-07-29	2025-07-13	3	
	subscription_type		average_watch_hours		mobile_app_usage_pct		\
0	2		42.6		77.4		
1	0		65.3		98.0		
2	1		40.1		47.8		
3	1		5.8		53.2		
4	2		32.7		16.8		
	complaints_raised		received_promotions		referred_by_friend		is_churned \
0	1.0		0		0		1.0
1	4.0		0		1		1.0
2	0.0		0		1		1.0
3	1.0		1		1		1.0
4	5.0		0		1		0.0
	monthly_fee	tenure_days	is_loyal	watch_per_ratio	heavy_mobile_user		\
0	10.99	159	False	0.267925	1		
1	5.99	893	True	0.073124	1		
2	13.99	1057	True	0.037938	0		
3	13.99	668	True	0.008683	0		
4	9.99	715	True	0.045734	0		
	low_watch_time		promo_and_low_watch				
0	0		0				
1	0		0				
2	1		0				
3	1		1				
4	1		0				

Additional behavioural indicators were created to track customers receiving promotions and users with low watch time, both of which were expected to influence churn risk.

These engineered features transformed transactional and behavioural data into measurable indicators of customer engagement and retention likelihood.

Key Business Insights

	feature	chi2	p_value	degrees_of_freedom	significant
0	gender	4.462637	0.107387	2	False
1	received_promotions	12.148069	0.002302	2	True
2	referred_by_friend	1.259159	0.532816	2	False

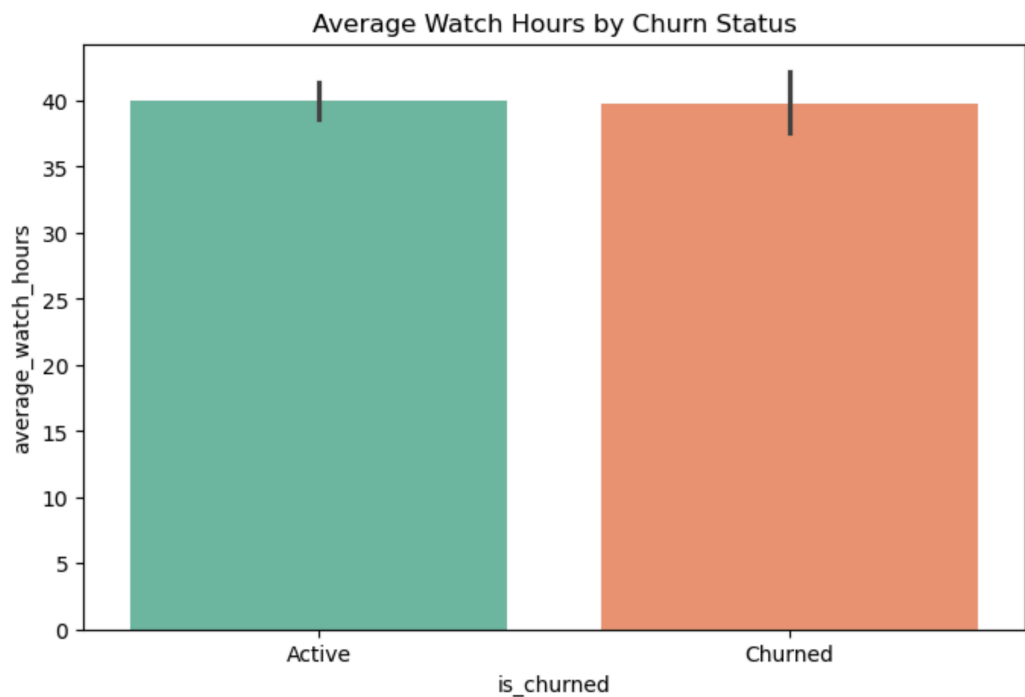
Promotions Significantly Influence Retention

One of the strongest findings from the analysis was the relationship between promotions and customer retention.

Chi-square testing revealed that users receiving promotions were significantly less likely to churn compared to those who did not receive promotional offers. This suggests that targeted promotions can be an effective retention strategy when applied correctly.

However, the analysis also indicated that promotions may have been directed toward already at-risk customers, highlighting correlation rather than direct causation.

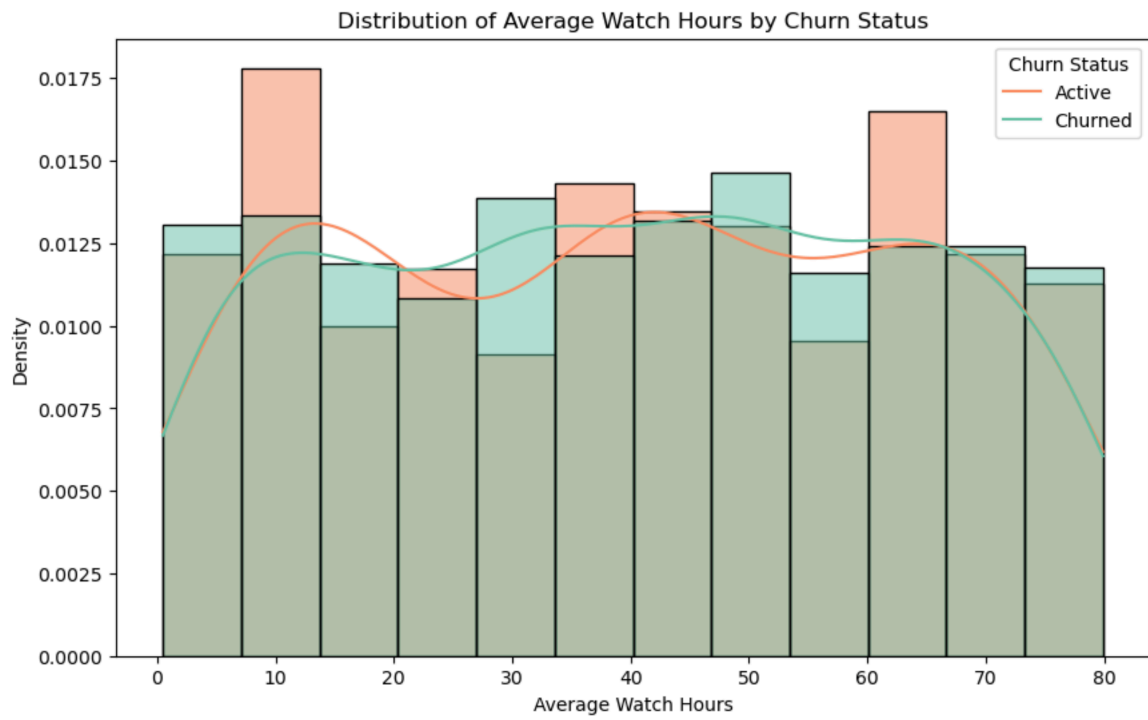
Watch Time Alone Does Not Predict Churn



Contrary to expectations, average watch time showed no statistically significant difference between churned and retained users.

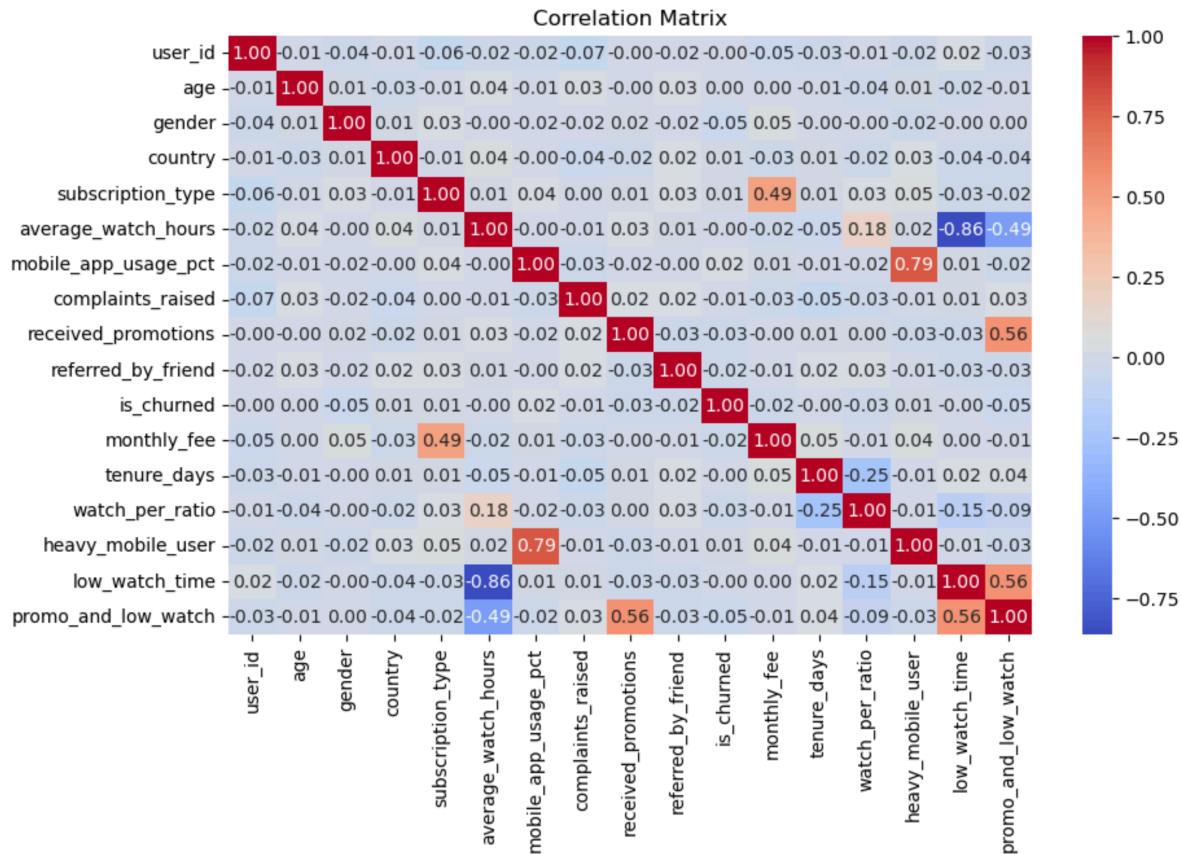
A t-test produced a high p-value, suggesting that viewing hours alone are not a reliable predictor of churn behaviour.

T-statistic: -0.14600709892499122
P-value: 0.8839677034293572
✗ No significant difference: watch time is similar between churned and retained users.



This finding indicates that customer retention is influenced more by perceived value and engagement quality than by raw consumption time.

Engagement and Loyalty Drive Retention



The analysis revealed three primary drivers of churn:

- Low watch time
- Low watch-to-fee ratio
- Short subscription tenure

Customers who engaged less with the platform or perceived lower value from their subscription were significantly more likely to cancel.

In contrast, loyal long-term subscribers demonstrated stronger retention patterns and higher engagement levels.

Mobile Usage Has Minimal Impact on Churn

Heavy mobile users displayed only a slightly higher churn rate compared to non-mobile users.

The difference was relatively small, suggesting that mobile usage itself is not a major factor driving cancellations. However, improving the mobile viewing experience could still contribute positively to long-term engagement.

Predictive Model Performance

A Logistic Regression model was developed to predict customer churn probability.

Logistic Regression Accuracy: 0.77

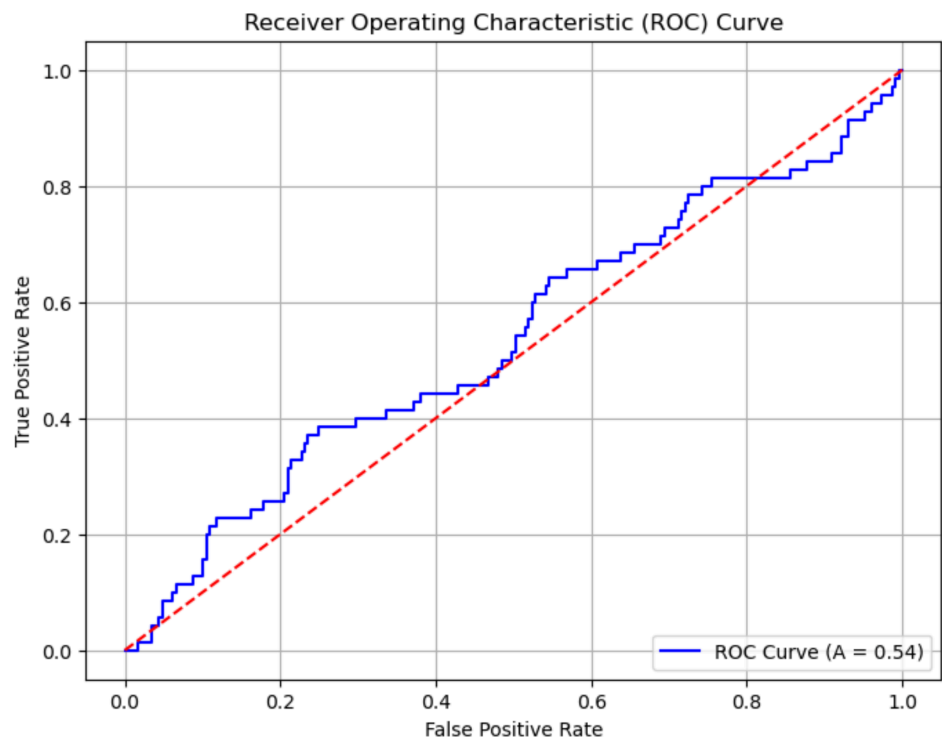
The model identified several influential churn indicators

Important Features for Churn Prediction:

	feature	importance
13	watch_per_ratio	0.177957
16	promo_and_low_watch	0.146312
12	is_loyal	0.136278

Feature	Importance
Watch-to-Fee Ratio	Indicates perceived customer value
Promotions + Low Watch Time	Identifies disengaged at-risk users
Loyalty Status	Loyal customers showed lower churn likelihood

Although these features contributed useful insights, the model’s ROC-AUC score indicated limited predictive power, suggesting the need for more advanced modelling approaches and additional behavioural data.



This highlights an important business insight: customer churn is complex and influenced by multiple interconnected behavioural patterns rather than a single variable.

Business Impact

The findings demonstrate that customer retention at StreamWorks Media depends heavily on customer engagement, perceived subscription value, and loyalty-building strategies.

Rather than relying solely on promotions or viewing hours, the business should focus on creating stronger long-term relationships with subscribers through personalised experiences, proactive engagement, and improved customer satisfaction.

Strategic Recommendations

1. Re-Engage Low Activity Users

Customers with low watch time and low engagement levels should be prioritised through:

- personalised recommendations,
- exclusive content alerts,
- targeted email campaigns,
- and in-app engagement reminders.

Improving engagement early can reduce churn risk significantly.

2. Expand Loyalty Programs

Loyal customers consistently demonstrated higher retention and stronger platform engagement.

StreamWorks Media should strengthen loyalty initiatives through:

- early content access,
- subscription rewards,
- loyalty discounts,
- and exclusive member benefits.

3. Improve Mobile Experience

Although mobile usage was not a major churn driver, enhancing the mobile app experience may improve long-term customer satisfaction.

Potential improvements include:

- smoother streaming performance,
- improved interface design,
- personalised mobile recommendations,
- and gamified engagement features.

Data Risks & Limitations

Several limitations were identified during the analysis:

- Dataset imbalance may reduce prediction reliability
- Some features risk data leakage by capturing post-churn behaviour
- Correlated behavioural features may introduce multicollinearity
- Linear models may oversimplify complex customer behaviour
- Insights may not generalise across all markets or time periods

Recognising these limitations is essential for improving future model accuracy and business decision-making.

Conclusion

This project demonstrates how customer churn is shaped less by demographics or watch time alone and more by perceived value, loyalty, and behavioural engagement.

Through data cleaning, feature engineering, statistical analysis, and predictive modelling, the analysis uncovered actionable insights that can help StreamWorks Media improve customer retention, reduce churn risk, and strengthen long-term profitability in an increasingly competitive streaming market.